Algebra 1
Final Exam

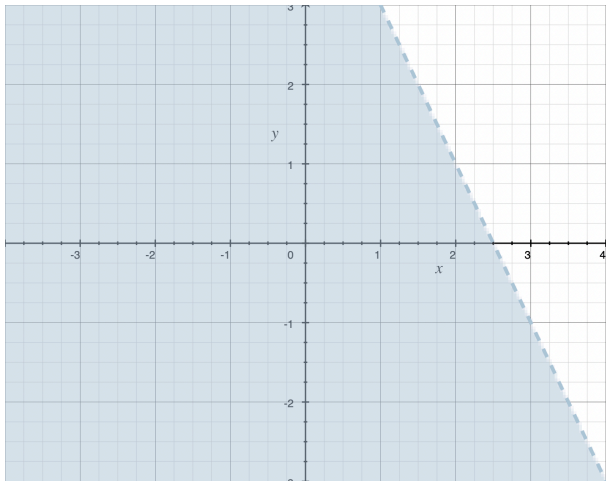
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Algebra 1
Final Exam Solutions

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Algebra 1 Final Exam Answer Key

1. (5 pts) A B C D **E**
2. (5 pts) **A** B C D E
3. (5 pts) A B **C** D E
4. (5 pts) A B C D **E**
5. (5 pts) A **B** C D E
6. (5 pts) A B **C** D E
7. (5 pts) A B C **D** E
8. (5 pts) A B C **D** E
9. (15 pts) $\dfrac{3x-3}{(x+5)(x-3)}$
10. (15 pts) $(3,1/2)$
11. (15 pts) $x=-16$ and $x=2$
12. (15 pts)



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Algebra 1 Final Exam Solutions

1. E. Apply the distributive property by multiplying the fraction $2x/yz$ by both fractions inside the parentheses.

$$\frac{2x}{yz}\left(\frac{4x}{3y}-\frac{x^2}{z}\right)=\frac{8x^2}{3y^2z}-\frac{2x^3}{yz^2}$$

2. A. The Commutative Property tells us that, when we're multiplying, we can "commute" the terms and move them around in any order.

3. C. "Three less than" means that we'll be subtracting 3 from whatever follows, which in this case is "the quotient of six and a number." "Quotient" means to divide, so we'll subtract 3 from $6/x$. The mathematical translation of this expression is

$$\frac{6}{x}-3$$

4. E. To solve, begin by distributing on both sides of the equation.

$$-(4x-10)+14=2(4x-6)$$

$$-4x+10+14=8x-12$$

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Combine like terms on the left.

$$-4x + 24 = 8x - 12$$

Add $4x$ and 12 to both sides to isolate x.

$$-4x + 4x + 24 + 12 = 8x + 4x - 12 + 12$$

$$36 = 12x$$

Divide both sides by 12.

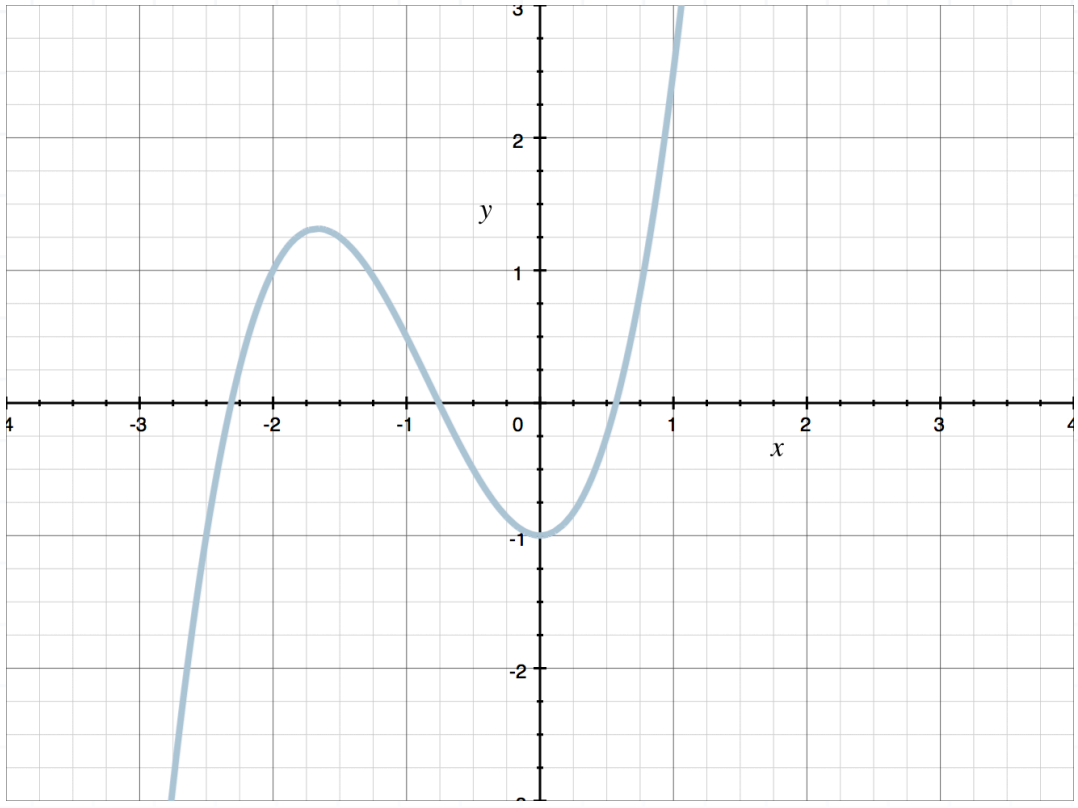
$$\frac{36}{12} = \frac{12x}{12}$$

$$3 = x$$

5. B. We can plug any x-value into the function $f(x) = 3\sqrt{x}$, except for negative integers, because the square root of any negative integer is undefined. Therefore the domain is $x \geq 0$.

6. C. Any graph that passes the Vertical Line Test represents a function. Answer choice C is the only graph where no vertical line will cross the graph more than one, which means the graph will only ever have one output y for any input x, and the graph therefore represents a function.

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7. D. Subtract 5 from both sides of the inequality.

$$5 - 4x \leq 13$$

$$5 - 5 - 4x \leq 13 - 5$$

$$-4x \leq 8$$

Divide both sides by -4. Since we're dividing by a negative, remember to switch the direction of the inequality from $\leq$ to $\geq$.

$$\frac{-4x}{-4} \geq \frac{8}{-4}$$

$$x \geq -2$$

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8. D. To evaluate $(fg)(-3)$, we can first find $f(-3)$ and $g(-3)$ and then find the product. With $f(x) = x - 3$ and $g(x) = 2x + 5$, we get

$$f(-3) = -3 - 3 = -6$$

$$g(-3) = 2(-3) + 5 = -1$$

Then the product is

$$(fg)(-3) = (-6)(-1) = 6$$

9. Find the LCD by factoring the denominator of the second fraction.

$$\frac{3}{x+5} + \frac{6}{x^2 + 2x - 15}$$

$$\frac{3}{x+5} + \frac{6}{(x+5)(x-3)}$$

Multiply the first fraction by $(x-3)/(x-3)$ so that both fractions have the same denominator, then add the fractions.

$$\frac{3}{x+5}\left(\frac{x-3}{x-3}\right) + \frac{6}{(x+5)(x-3)}$$

$$\frac{3(x-3)}{(x+5)(x-3)} + \frac{6}{(x+5)(x-3)}$$

$$\frac{3x-9}{(x+5)(x-3)} + \frac{6}{(x+5)(x-3)}$$

Combine the fractions by adding the numerators.

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$$\frac{3x-3}{(x+5)(x-3)}$$

10. We can solve using substitution. Substitute $6-6y$ for x into the first equation, then solve for y.

$$4x+6y=15$$

$$4(6-6y)+6y=15$$

$$24-24y+6y=15$$

$$24-18y=15$$

$$24=15+18y$$

$$9=18y$$

$$y=\frac{1}{2}$$

Plug $y=1/2$ into the second equation to solve for x.

$$x=6-6\left(\frac{1}{2}\right)$$

$$x=6-3$$

$$x=3$$

The single solution is the point $(3,1/2)$.

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11. Find the factors of -32 that add to 14. One factor will be negative.

$$-16+2=-14$$

$$-2+16=14$$

$$-8+4=-4$$

$$-4+8=4$$

Since $-2+16=14$, the factors of $x^2+14x-32$ are

$$(x-2)(x+16)$$

Set each factor equal to 0 and solve for x.

$$x-2=0$$

$$x=2$$

and

$$x+16=0$$

$$x=-16$$

The solutions are $x=2$ and $x=-16$.

12. Start by solving for y.

$$-x-y>x-5$$

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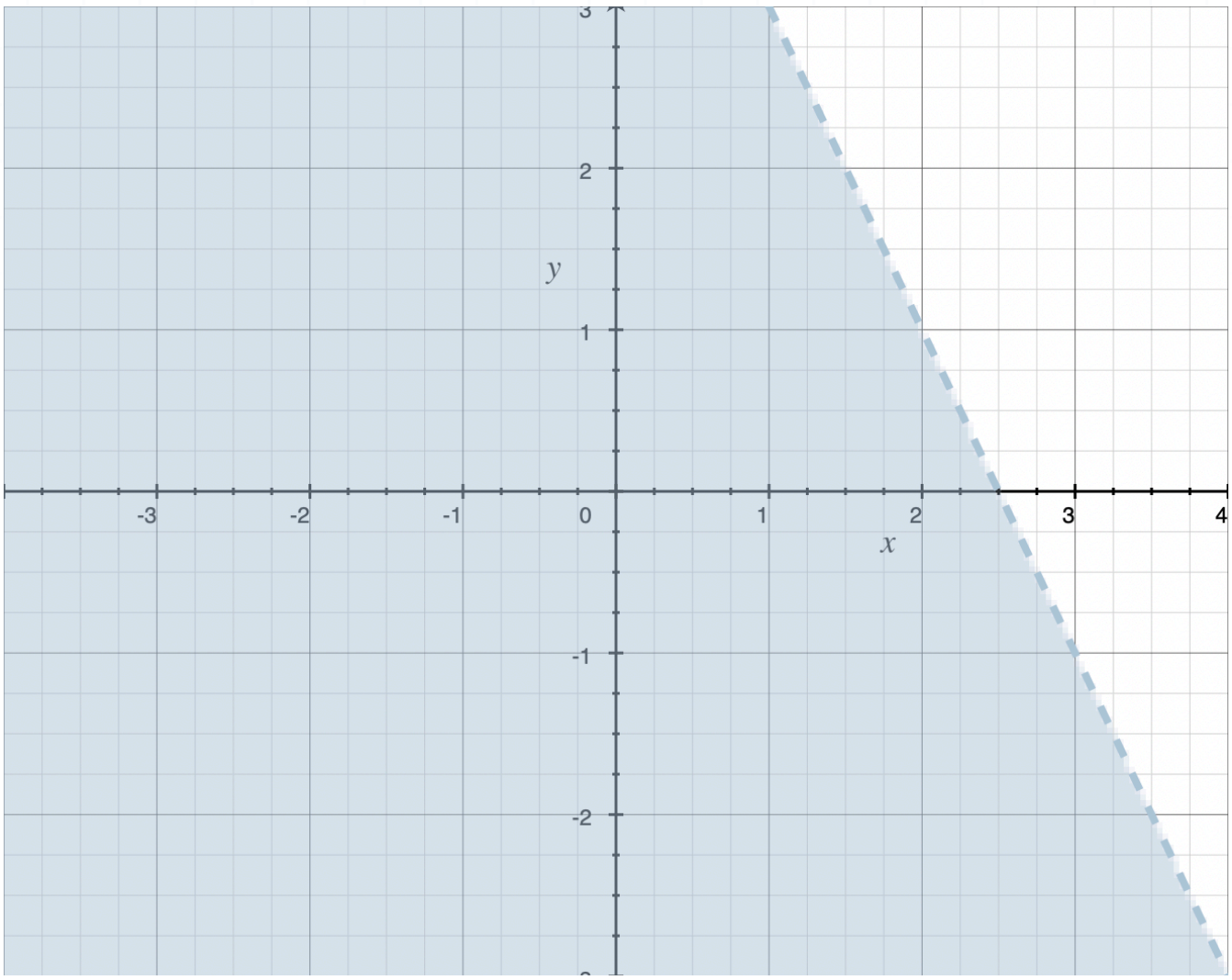
$$-x + x - y > x + x - 5$$

$$-y > 2x - 5$$

$$-1 \cdot -y > -1(2x - 5)$$

$$y < -2x + 5$$

Graph the line starting with a point on the y-axis at 5. Find the next point by using the slope and going down 2 and to the right 1. Make sure to draw a dotted line and shade "under" the line to the left, since the inequality is strictly less than y.



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